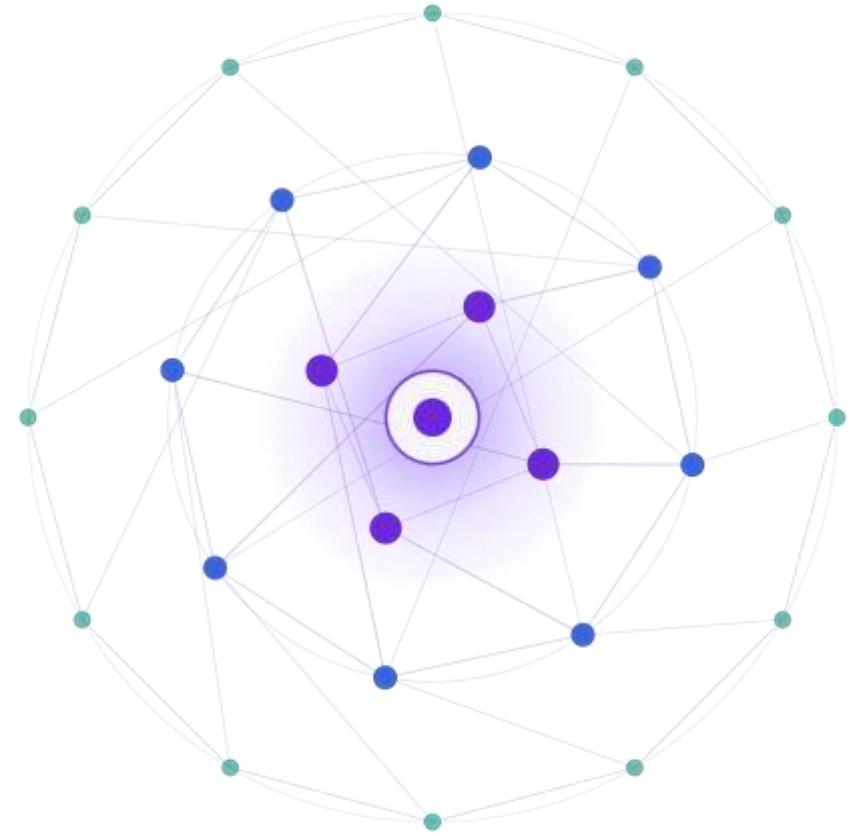


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 05
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- Submission Github link -
<https://github.com/joannamariyajames/interjectsamsung>



Theme

THEME 05 · INTERRUPTIBLE REAL-TIME AGENTS

People don't talk in turns. They pause, change their minds and cut in.

A voice agent has to keep up — without going silent, redoing work or acting on what the user took back.

1

Stay responsive

Acknowledge at once. Never go silent while thinking, and never say something is done before it is.

2

Work asynchronously

Start retrieval and tools while the user is still speaking. Stream the answer and keep it interruptible.

3

Recover cleanly

When cut off or corrected, drop the stale work, keep what is still useful, and never run an action twice.

Measured on Full-Duplex-Bench v3: 100 real recordings full of pauses, fillers, hesitations, false starts and self-corrections, across 4 domains (e-commerce, finance, housing, travel) and 12 tools. The agent must call the right tool with the user's final intent.

Existing Solutions & Gaps

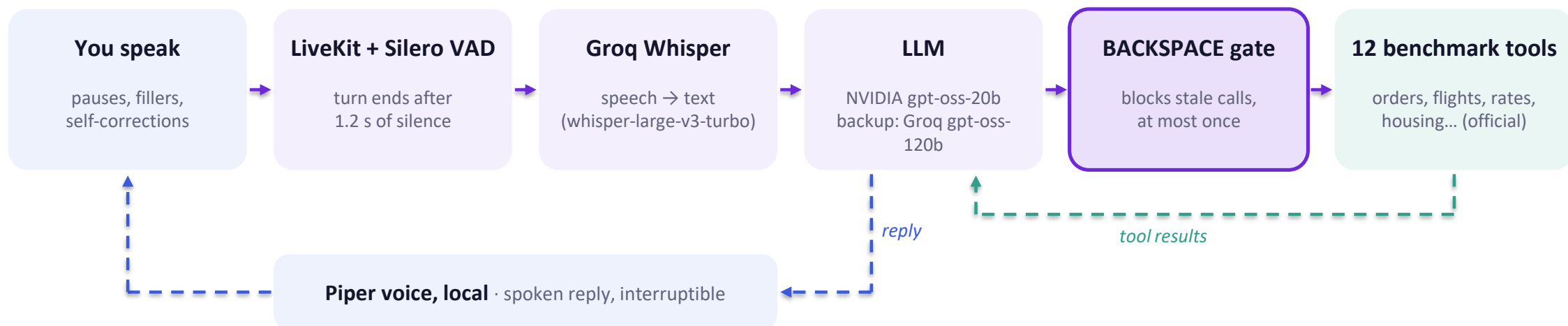
Today's voice agents are built for turn-taking. Interruptions are treated as noise, not as a signal.

Approach	How it works	Gap
Turn-based voice pipelines	Wait for silence → speech-to-text → LLM → speech	A mid-sentence pause ends the turn early, so the agent acts on half a request. Dead air while the model thinks. Barge-in just stops the audio and throws the work away.
Speech-to-speech realtime models	One model hears and speaks directly	Usually paid and closed. Little control over when tools fire, so a correction can still trigger the stale action. Hard to reproduce a score.
Chat assistants with tools	Text in, tool calls out	Act on the first value heard. “Agra... actually no, Jaipur” can leave an Agra booking behind, or repeat an action twice.

The missing piece: treat the interruption itself as a first-class event — stop at once, keep the useful work, drop what was superseded, and make every action happen at most once.

Our Solutions & Architecture Diagram

Interject: a LiveKit voice agent whose every tool call passes through BACKSPACE, our interruption-aware gate.



BACKSPACE gate

- A call whose inputs were superseded by a correction is blocked before it runs
- Every action is committed at most once
- Values normalised: spelled codes joined, dates kept as spoken

Built to stay up

- Backup LLM takes a turn when NVIDIA errors or stays silent for 10 s
- Turn ends after 1.2 s of silence, so a pause is not a finished request
- Every call logged for the official evaluator

Same core in the web app

- Retrieval starts on partial speech, before you finish
- Cut in → turn cancelled, checkpoint kept; “go on” resumes
- Goal stack parks a detour; a harness refuses unsafe tools

Demo & Product Walkthrough

Two live surfaces: the assistant you can talk over, and Drive, an in-car assistant built on the same core.

The image displays three screenshots from a demo. The left screenshot shows a chat interface with a user asking "What are the baggage limits on each cabin?" and a system response about Interject Travel's baggage policy. The middle screenshot shows a "DRIVE" interface with performance metrics like "AVERAGE TIME TO YIELD: 0.00 ms" and "LATENCY REMOVED: 260 ms". The right screenshot shows a "WHAT THE AGENT DID" log with steps like "SPEC Started planning the route to Jaipur, Rajasthan" and "NAV STARTED Navigation started to Jaipur, Rajasthan".

Assistant: cut in mid-answer → it stops at once (0.00 ms average time to yield), holds a checkpoint, and “go on” continues from the exact words.

Drive

“Take me to, um, Agra... actually no, Jaipur”

Plans Agra while you speak, drops it on the correction, and starts navigation once — to Jaipur.

Demo video (under 5 min): <https://www.youtube.com/watch?v=UI59hoOJY4A>

Tools and tech stack used

Everything runs on free tiers or locally — no paid API or service anywhere in the stack.

Voice agent (benchmark)

LiveKit Agents 1.8.3

rooms, turn handling, fallback

Silero VAD

voice activity, local

Groq whisper-large-v3-turbo

speech-to-text

NVIDIA gpt-oss-20b

tool-calling LLM, seed 7, temp 0

Groq gpt-oss-120b

backup LLM

Piper en_US-ljspeech-medium

local voice, SHA-256 pinned

Web app & Drive

Python 3.11 · FastAPI

WebSocket streaming backend

React · Vite · Zustand

live UI

Tailwind CSS v4

light & dark themes

Web Speech API

mic + voice in Chrome/Edge

SQLite + scrypt

accounts, hashed sessions

Groq gpt-oss-120b

general answers (optional)

Evaluation & quality

Full-Duplex-Bench v3

pinned commit, official scripts

run_fdb_v3.sh

one command: install → run → score

pytest

871 backend tests

Playwright

7 real-browser suites

GitHub Actions

CI on every push

Pinned versions

requirements + seed recorded

Impact & Use case

Where interruptions matter most: a driver who can't look at a screen and changes plans mid-sentence.

Use case: Drive — an in-car voice assistant for India

- **Change of mind mid-sentence**
“Agra... actually no, Jaipur” → acts only on Jaipur
- **“Hold on” / “go on”**
pauses on request, resumes from where you stopped listening
- **Add stops on the way**
the old route goes stale and is recomputed with the stop
- **Ask twice, start once**
a repeated request never starts navigation a second time
- **Works offline**
no map service or API key needed for the demo

Safer hands-free use

The driver corrects by voice; no need to look at or tap a screen to undo a mistake.

No double or stale actions

At-most-once commits: no second booking, payment or navigation from a repeated or corrected request.

Beyond the car

Same gate works for support, banking, travel and housing — the four domains of the benchmark.

Innovation highlights, results and limitations

56%

strict pass rate

all 100 recordings, audio via LiveKit

99%

turn-taking

the agent answered 99 of 100

97%

right tool chosen

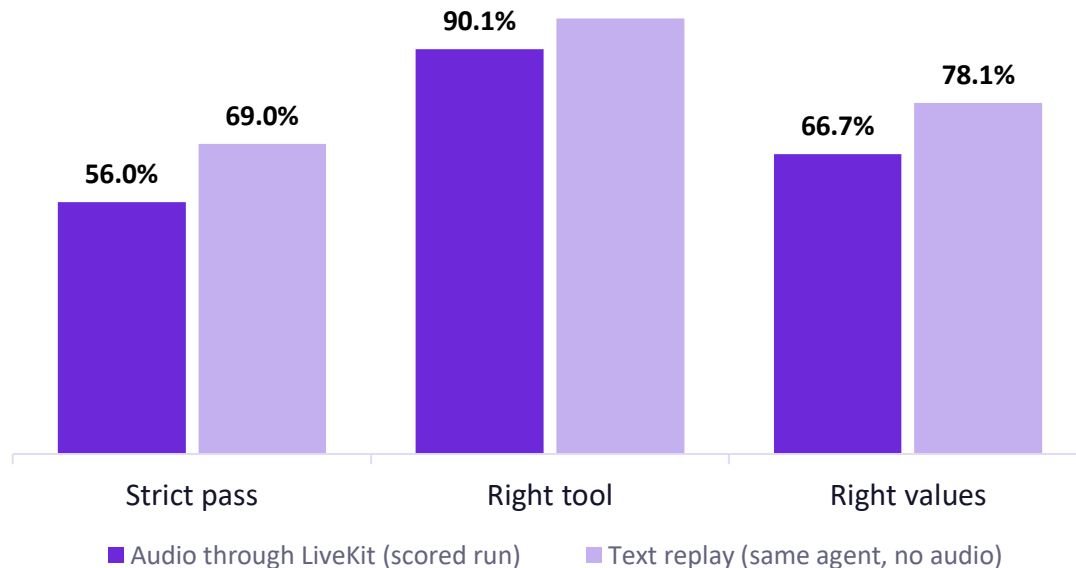
tool selection accuracy

4.45 s

average latency

end of speech → reply

Full-Duplex-Bench v3, 100 recordings (%)



Innovation highlights

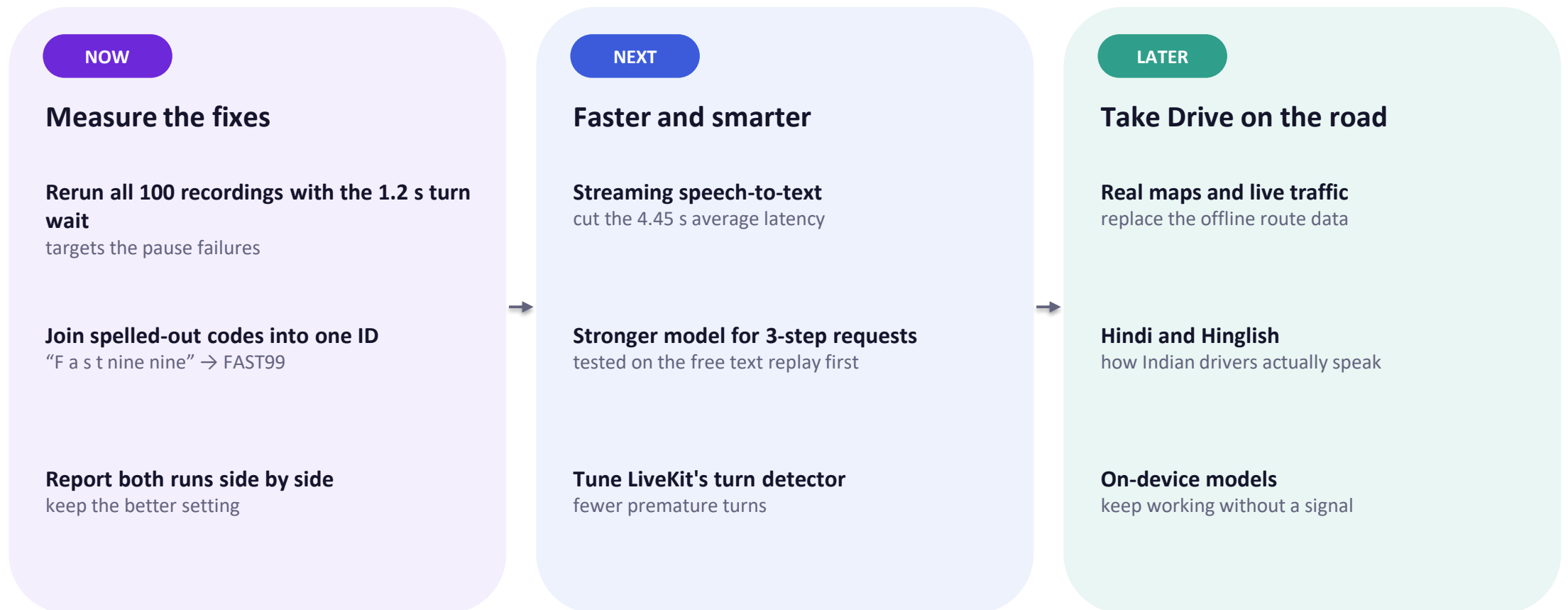
- BACKSPACE gate: stale tool calls blocked before they run
- Backup LLM answered 12 of 384 requests while NVIDIA was failing
- Speculative retrieval + checkpoint resume; time to yield < 1 ms

Limitations (and what caused them)

- Pauses read as the end of a request: most of the 16 audio-only failures
- Spelled codes misheard ("F a s t nine nine" for FAST99)
- 3-step requests pass 25%; average latency 4.45 s

What's next

Each next step comes straight from a measured failure, so each one can be checked with the same benchmark.



Brownie points slide(differentiation)

What sets Interject apart: it is free to reproduce, tested end to end, and honest about what it knows.

₹0

to reproduce

Every provider is a free tier or local. One command installs, runs and scores the benchmark.

871

backend tests

Plus 7 real-browser suites and CI on every push.
No test weakened to pass.

12

backup answers

In the full run the backup model answered 12 of 384 requests while NVIDIA's free endpoint was failing.

< 1 ms

to stop talking

A barge-in cancels the turn at once, and the checkpoint keeps what was already said.

0

hardcoded answers

A test proves the agent's rules never quote benchmark data; fixes are general rules.

Honest

by design

Offline mode says so, the demo knowledge base is marked fictional, and no action is claimed unless it ran.

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N) **Y**
- README with reproducible setup instructions (Y/N) **Y**
- Demo video, max 5 minutes (YouTube or Drive link)
<https://www.youtube.com/watch?v=UI59hoOJY4A>
- Presentation file (PPT or PDF) (Y/N) **Y**

Repository: github.com/joannamariyajames/interjectsamsung

README: judges' quick start (no keys needed for the app) and a one-command benchmark run.

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India